

Analytical solution for the problem of one-dimensional quasi-steady non-charring ablation in a semi-infinite solid with temperature-dependent thermo-physical properties

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ABSTRACT

Exact quasi-steady solution of temperature distribution for one-dimensional non-charring ablation in a semi-infinite material with temperature-dependent thermo-physical properties is obtained analytically, in which the surface temperature and the surface recession velocity are assumed to be constant. The analytical solutions for the cases in which thermo-physical properties are expressed by low degree polynomial functions of temperature are also obtained. Two kinds of non-charring ablation models are considered. One is a single phase model composed of a single phase (the crystalline phase), in which the surface recedes due to ablation. The other is a two-phase model composed of two phases (the crystalline and the amorphous phases) such as Teflon, in which the surface of the amorphous region recedes. The calculated results of the analytical solution agreed well with those of the numerical solution by the finite difference method. The basic behavior of Teflon ablation is calculated by using the analytical solution. The simple and exact analytical solution obtained in this paper is easy to calculate and is helpful in understanding the ablation behavior easily. It allows for rapid estimation of the ablation behavior for engineering purposes. The solution also provides a means to verify computer solutions obtained by numerical methods.

1. Introduction

The term ablation in this paper refers to mass loss due to thermochemical processes, together with an associated removal of heat in high heating environments. Applications involving ablation include thermal protection of atmospheric re-entry vehicles, nozzles of rocket engines, etc. When high heating is applied on an ablative material, the temperature of the material becomes high and ablation processes such as sublimation, melting, vaporization, oxidation and chemical decomposition (depolymerisation, pyrolysis) of the material occur, which reduce heat flow to the inside. The mechanisms of ablation differ according to various kinds of ablative materials [1–3]. Examples of ablative materials are carbon phenolic, Teflon, graphite, and Pyrex. The carbon phenolic ablator [4] is classified as a charring ablator, in which at high temperatures the phenolic resin decomposes in-depth, yielding the pyrolysis gases that percolate to the surface and leave behind a carbon residue on the carbon cloth. Ablators made of Teflon, graphite, and Pyrex are non-charring ablators, in which ablation occurs due to thermochemical processes, such as sublimation, melting, vaporization and oxidation, leaving no charred materials.

Ablation analysis plays an important role in the design of ablators. Until now, various numerical methods of ablation analysis have been developed and a lot of numerical simulations have been carried out [4–14]. Potts [4] reviewed application of HBI (Heat Balance Integral) methods to realistic models of material response, specifically, for carbon-carbon and carbon-phenolic heat shields on reentry vehicles. Moyer and Rindal [5] developed a one-dimensional finite difference ablation code. This code solved the internal energy balance and decomposition equations, coupled with the ablating surface energy balance condition, to simulate the response of ablative heatshields in hypersonic flows. Milos and Chen [6] described and demonstrated new capabilities in the two-dimensional implicit thermal response and ablation program. Kato et al. [7] conducted thermal response simulation of ultra light weight phenolic carbon ablator (LATS) by using the ablation analysis code. The agreement between the results of simulation and measurement by arc-heated tests was found to be good. Kato et al. [8] examined the effects of density, thickness and heat load upon the heat shield performance of the lightweight phenolic carbon ablator named LATS quantitatively for both arc-heated tests and re-entry conditions using a one-dimensional ablation analysis code. It was found that the necessary thickness is nearly constant for various ablator densities. Kato

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Nomenclature			
C_p	specific heat	T_s	surface temperature
C_{p0}, C_{p1}, C_{p2}	coefficients for the temperature function of specific heat	T_0	initial temperature
FDM	Finite Difference Method	V	velocity of surface recession
h_m	latent heat of phase transition from the crystalline (solid) to the amorphous (gel) phase	V_m	velocity of phase transition plane
$I_B(T_s, T), I_C(T_s, T)$	Eq. (2.3-5)	x	moving coordinate with the origin O' at moving surface
$I_B(T_m, T), I_C(T_m, T)$	Eq. (3.3-4)	y	stationary coordinate with the origin O at initial surface
k	thermal conductivity	z	moving coordinate with the origin O'' at phase transition plane P_m
k_0, k_1, k_2	coefficients for the temperature function of thermal conductivity	Greek letters	
L_∞	x or z coordinate of boundary point for the calculation of transient heat transfer by FDM	α	a real number and one of the roots for the cubic equation $a_3v^3 + a_2v^2 + a_1v + a_0 = 0$
P_m	plane of phase transition from the crystalline (solid) to the amorphous (gel) phase	δ_m	thickness of amorphous region
O, O', O''	origins of y, x , and z axes, respectively	ρ	density
Q	heat flux	ρ_0, ρ_1, ρ_2	coefficients for the temperature function of density
S	recession length	ρ_m	density at phase transition plane ($= (\rho_a + \rho_c)/2$)
t	time	(ρ_a, ρ_c)	densities of the amorphous and the crystalline phases, respectively
T	temperature	Subscript	
T_m	phase transition temperature at phase transition plane	s	surface
		0	initial
		m	phase transition

et al. [9] investigated the coking phenomenon within a lightweight carbon-phenolic ablator exposed to the heating environment of air by using the ablation analysis code which can calculate the coking behavior. The effect of coking upon the ablation behavior was studied analytically and experimentally. Clark [10] calculated the transient ablation of Teflon layers (two-layer model) numerically by the finite difference method. It was concluded that the ablation model accurately predicted the transient thermal response and surface recession of ablating Teflon. Holzknicht [11] investigated the high-temperature behavior of Teflon. An analytical model for the transient one-dimensional ablation of Teflon layers (two-layer model) was developed. The crystalline-amorphous phase transition, the thermal expansion and the formation of higher-molecular products at the surface were taken into account. Comparisons of computed temperature profiles with measured values in plasma-arc jet facilities confirmed the model for transient ablation and the limiting case of quasi-steady ablation. Arai [12] conducted a thermal analysis for transient response of ablating materials made of high molecular compounds on the basis of two-layer thermal model by use of an appropriate numerical method. Numerical computation was carried out for an ablating blunt body of revolution made of Teflon. Comparison of the numerical results with the experimental data revealed a remarkable fact that the single-layer thermal model does not predict the real feature of the transient thermal field, while the two-layer thermal model agrees well with the experiment. Arai [13] conducted the transient one-dimensional ablation analysis of Teflon, in which the optical transmittance of both the amorphous zone and the crystalline zone of Teflon-layer was taken into account. The results of the calculation confirm that the optical transmittance of both the amorphous zone (gel layer) and the crystalline zone cannot be neglected in considering the transient ablation under the condition of the radiative heating. Nomura [14] conducted measurements of transient ablation of Teflon. Quantitatively good agreement was obtained between both the results of experiment and a finite difference analysis of heat conduction in Teflon, in which the melting layer having considerably smaller thermal conductivity than that of solid Teflon was taken into account.

As for the analytical solutions for ablation problems, only a few have been investigated in the literature. If the simple and exact analytical solution for a simple ablation problem is available, it would be easy to

calculate and will be helpful in understanding the ablation behavior easily. Landau [15] gave the exact quasi-steady solution for a semi-infinite melting solid with its face heated at a constant rate. The exact transient solution for a large latent heat was also obtained. Thermal properties of the solid are assumed constant. The liquid is assumed to be removed immediately on formation so that the face of the solid moves inward. Bear and Ambrosio [16] derived exact solutions of temperature in which sublimation occurs at the surface of a semi-infinite slab. By using the method of Laplace Transformation exact solutions of steady state and transient conditions for constant sublimation rate with constant thermal properties were obtained. Lin [17] investigated the quasi-steady ablation process on the surface of a two-layer composite consisting of a layer of ablation material combined with another layer of non-ablative substrate theoretically. Exact solutions of the quasi-steady ablation problem were derived. Quan [18] derived the first-order approximate solution in analytic form for steady state charring ablation in a semi-infinite ablator. Quasi-steady temperature and density distributions in the ablator were obtained.

As for analytical solution for one-dimensional steady-state non-charring ablation with constant thermal properties in a semi-infinite solid, it is well known [5,15,19]. However, thermo-physical properties of ablative materials are temperature-dependent. If analytical solution with temperature-dependent thermo-physical properties is available, it would be very useful. As far as the authors know, analytical solution of non-charring ablation with temperature-dependent thermo-physical properties has not been treated in the literature.

In this paper, quasi-steady solution for one-dimensional non-charring ablation in a semi-infinite material with temperature-dependent thermo-physical properties is derived analytically, where the surface temperature and the surface recession velocity are assumed to be constant. The solutions for the cases in which thermo-physical properties are expressed by low degree polynomial functions of temperature are also derived. Two kinds of non-charring ablation models are considered for the problem of quasi-steady ablation. One is a single phase model composed of a single phase (the crystalline (solid) phase), in which the surface recedes due to ablation. The other is a two-phase model made of two phases (the crystalline (solid) and the amorphous (gel) phases) such as Teflon, in which the surface of the amorphous region recedes. The validity of the analytical solution is discussed. The ablation behavior of

Teflon is examined by calculating the analytical solution.

2. Non-charring ablator composed of a single phase

Analytical solution for quasi-steady one-dimensional non-charring ablation in a semi-infinite material with a single phase (the crystalline (solid) phase) is investigated. In the quasi-steady-state condition, the surface temperature is assumed to be a constant. The surface is assumed to recede with a constant velocity due to ablation. Phase change does not occur inside the ablative material. The thermal properties are assumed to be temperature-dependent.

2.1. Ablation model and basic equations

Consider a one-dimensional semi-infinite solid made of a non-charring ablative material of a single phase (a single phase model), as shown in Fig. 1. The solid is assumed initially at a constant temperature T_0 , and then suddenly a constant heat flux Q is loaded on the front surface. If the value of Q is high enough and heating continues long enough, the surface recedes at a constant velocity V and the surface temperature keeps a constant value T_s . It is assumed that the material consists of a single phase (the crystalline (solid) phase) and ablation occurs at the surface. The axes of x and y are considered: x represents the moving coordinate in which the origin O' is fixed at the moving surface, and y represents the stationary coordinate in which the origin O is fixed at the initial surface. The length of OO' is $S(t)$. The thermal conductivity k and the specific heat C_p of the material are assumed to be functions of temperature T . The density ρ is assumed to be a constant. In this model thermal expansion is neglected [10,13].

The one-dimensional unsteady heat conduction equation (the energy equation) in a semi-infinite solid in the stationary coordinate system is given as

$$\rho C_p \frac{\partial T}{\partial t} \Big|_y = \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right)_t \quad (2.1-1)$$

Here, we transform this equation to an equation in the moving coordinate system, following the procedure by Mayer and Rindal [5]. T is a function of y and t , $T = T(y, t)$. Then

$$dT = \frac{\partial T}{\partial y} \Big|_t dy + \frac{\partial T}{\partial t} \Big|_y dt \quad (2.1-2)$$

Differentiating with respect to time t at constant x yields

$$\frac{\partial T}{\partial t} \Big|_x = \frac{\partial T}{\partial y} \Big|_t \frac{\partial y}{\partial t} \Big|_x + \frac{\partial T}{\partial t} \Big|_y \quad (2.1-3)$$

From Fig. 1, the x and y coordinates are related as

$$y = x + S(t) \quad (2.1-4)$$

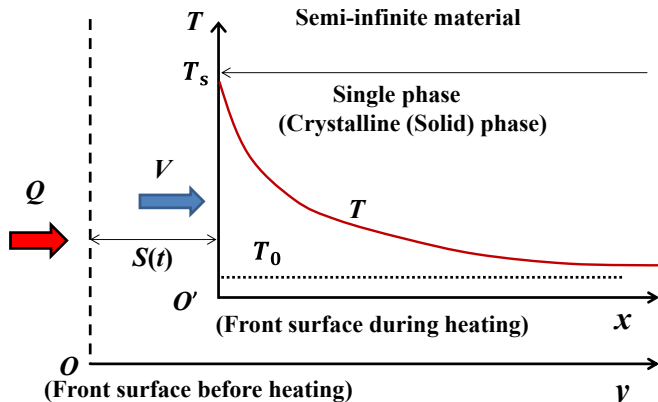


Fig. 1. Semi-infinite non-charring ablator (Quasi-steady single phase model).

from which

$$\frac{\partial y}{\partial t} \Big|_x = \frac{dS}{dt} \quad (2.1-5)$$

Apparently we have

$$\frac{\partial T}{\partial y} \Big|_t = \frac{\partial T}{\partial x} \Big|_t \quad (2.1-6)$$

Substituting Eqs. (2.1-5) and (2.1-6) into Eq. (2.1-3) yields

$$\frac{\partial T}{\partial t} \Big|_x = \frac{\partial T}{\partial x} \Big|_t \frac{dS}{dt} + \frac{\partial T}{\partial t} \Big|_y \quad (2.1-7)$$

Substitution of Eq. (2.1-7) into Eq. (2.1-1) with the observation that partial differentiation with respect to x or y at constant time is equivalent results in the transformed energy equation

$$\rho C_p \frac{\partial T}{\partial t} \Big|_x = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right)_t + \rho C_p \frac{dS}{dt} \frac{\partial T}{\partial x} \Big|_t \quad (2.1-8)$$

In the quasi-steady-state condition, the left side term of Eq. (2.1-8) becomes zero, dS/dt becomes a constant V , and T becomes the function of x only, then we obtain

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + \rho C_p V \frac{dT}{dx} = 0 \quad (2.1-9)$$

with the boundary conditions

$$T(0) = T_s \quad (2.1-10a)$$

$$T(x) \rightarrow T_0 \text{ as } x \rightarrow \infty \quad (2.1-10b)$$

Apparently from Eq. (2.1-10b) the following equation is also obtained.

$$dT(x)/dx \rightarrow 0 \text{ as } x \rightarrow \infty \quad (2.1-10c)$$

It is assumed that in Eqs. (2.1-9) to (2.1-10c), ρ , V , T_s and T_0 are constants, and k and C_p are functions of T .

As an example of application, the ablation model mentioned above can be applied to the situation in which the ablative material is under the heating test with constant Q . (If Q is constant, T_s and V are also constant in a quasi-steady state condition.)

2.2. Analytical solution for the single phase model

In this section, analytical solution for the boundary value problem of Eqs. (2.1-9) to (2.1-10c) is derived. Eq. (2.1-9) can be expressed as

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + V \frac{d}{dx} \int_{T_0}^T \rho C_p(u) du = 0 \quad (2.2-1)$$

Eq. (2.2-1) yields

$$\frac{d}{dx} \left[k \frac{dT}{dx} + V \int_{T_0}^T \rho C_p(u) du \right] = 0 \quad (2.2-2)$$

Then

$$k \frac{dT}{dx} + V \int_{T_0}^T \rho C_p(u) du = C \quad (2.2-3)$$

where C is a constant. Substitution of Eqs. (2.1-10b) and (2.1-10c) into Eq. (2.2-3) gives $C = 0$, then

$$\frac{dT}{dx} = -\frac{V}{k} \int_{T_0}^T \rho C_p(u) du \quad (2.2-4)$$

The reciprocal of Eq. (2.2-4) gives

$$\frac{dx}{dT} = -\frac{k(T)}{\int_{T_0}^T \rho C_p(u) du} \quad (2.2-5)$$

In Eq. (2.2-5) T is defined in $T_0 < T \leq T_s$. Integrating Eq. (2.2-5) and using Eq. (2.1-10a) yields the solution

$$x = \frac{1}{V} \int_T^{T_s} \frac{k(v)}{\int_{T_0}^v \rho C_p(u) du} dv \quad (2.2-6)$$

Apparently, the denominator in the integrand is positive for $T_0 < v \leq T_s$. This equation expresses the analytical solution of the boundary value problem of Eqs. (2.1-9) to (2.1-10c) for the single phase ablation model.

In Section 2.3, simple analytical solutions are derived when $k(T)$ and $C_p(T)$ are expressed by low degree polynomials of T , and ρ is a constant.

The ablation model in this section can be applied to the Teflon material (single phase model) which is made from a solid polymer. At the surface, sublimation is assumed to occur at elevated temperatures, where the material decomposes from the solid to the gas state.

2.3. Simple analytical solution

It is assumed that k , C_p and ρ of the ablative material are expressed by the following equations

$$\begin{aligned} k(T) &= k_0 + k_1 T + k_2 T^2, \\ C_p(T) &= C_{p0} + C_{p1} T + C_{p2} T^2, \\ \rho &= \rho_0 \end{aligned} \quad (2.3-1)$$

where $k(T)$, $C_p(T)$, and ρ are positive.

2.3.1. Case 2.3-1 (k : constant, linear or quadratic function of T , C_p : Quadratic function of T , ρ : constant)

Substituting Eq. (2.3-1) with $C_{p2} \neq 0$ into Eq. (2.2-6) yields

$$\begin{aligned} x &= \frac{1}{V} \int_T^{T_s} \frac{(k_0 + k_1 v + k_2 v^2)/\rho_0}{C_{p0}(v - T_0) + \frac{C_{p1}}{2}(v^2 - T_0^2) + \frac{C_{p2}}{3}(v^3 - T_0^3)} dv \\ &= \frac{1}{V\rho_0} [p \int_T^{T_s} \frac{1}{v - T_0} dv + q \int_T^{T_s} \frac{1}{Av^2 + Bv + C} dv + r \int_T^{T_s} \frac{v}{Av^2 + Bv + C} dv] \end{aligned} \quad (2.3-2a)$$

$$(2.3-2b)$$

where

$$\begin{aligned} T_0 < T \leq T_s, A &= \frac{C_{p2}}{3}, B = C_{p1}/2 + (C_{p2}/3)T_0, C = C_{p0} + (C_{p1}/2)T_0 + (C_{p2}/3)T_0^2 \\ p &= \frac{k_2 T_0^2 + k_1 T_0 + k_0}{AT_0^2 + BT_0 + C}, q = \frac{(Ck_2 - Ak_0)T_0 + Ck_1 - Bk_0}{AT_0^2 + BT_0 + C} \\ r &= \frac{(k_2 B - Ak_1)T_0 + Ck_2 - Ak_0}{AT_0^2 + BT_0 + C} \end{aligned} \quad (2.3-3)$$

Therefore we obtain the solution

$$x = \frac{1}{V\rho_0} (p \ln \frac{|T_s - T_0|}{|T - T_0|} + q I_B(T_s, T) + r I_C(T_s, T)) \quad (2.3-4)$$

where

$$\begin{aligned} I_B(T_s, T) &= \int_T^{T_s} \frac{1}{Av^2 + Bv + C} dv, \\ I_C(T_s, T) &= \int_T^{T_s} \frac{v}{Av^2 + Bv + C} dv \end{aligned} \quad (2.3-5)$$

When the discriminant D is defined by the following equation,

$$D = B^2 - 4AC \quad (2.3-6a)$$

$I_B(T_s, T)$ and $I_C(T_s, T)$ are given as follows [20].

(a) If $D > 0$ then

$$I_B(T_s, T) = \frac{1}{\sqrt{D}} \ln \frac{|2AT_s + B - \sqrt{D}|}{|2AT_s + B + \sqrt{D}|} - \frac{1}{\sqrt{D}} \ln \frac{|2AT + B - \sqrt{D}|}{|2AT + B + \sqrt{D}|} \quad (2.3-6b)$$

(b) If $D < 0$ then

$$I_B(T_s, T) = \frac{2}{\sqrt{-D}} \tan^{-1} \frac{2AT_s + B}{\sqrt{-D}} - \frac{2}{\sqrt{-D}} \tan^{-1} \frac{2AT + B}{\sqrt{-D}} \quad (2.3-6c)$$

(c) If $D = 0$ then

$$I_B(T_s, T) = \frac{-2}{2AT_s + B} + \frac{2}{2AT + B} \quad (2.3-6d)$$

$I_C(T_s, T)$ is expressed by

$$I_C(T_s, T) = \frac{1}{2A} \ln \frac{|AT_s^2 + BT_s + C|}{|AT^2 + BT + C|} - \frac{B}{2A} I_B(T_s, T) \quad (2.3-7)$$

2.3.2. Case 2.3-2 (k : constant, linear or quadratic function of T , C_p : Linear function of T , ρ : constant)

Substituting Eq. (2.3-1) with $C_{p2} = 0$ and $C_{p1} \neq 0$ into Eq. (2.2-6) yields

$$x = \frac{1}{V\rho_0} \int_T^{T_s} \frac{k_0 + k_1 v + k_2 v^2}{C_{p0}(v - T_0) + \frac{C_{p1}}{2}(v^2 - T_0^2)} dv \quad (2.3-8a)$$

$$= \frac{1}{V\rho_0} \int_T^{T_s} \frac{k_0 + k_1 v + k_2 v^2}{(v - T_0)\{C_{p0} + \frac{C_{p1}}{2}(v + T_0)\}} dv \quad (2.3-8b)$$

$$= \frac{1}{V\rho_0} \int_T^{T_s} [C_0 + \frac{C_1}{v - T_0} + \frac{C_2}{C_{p0} + \frac{C_{p1}}{2}(v + T_0)}] dv \quad (2.3-8c)$$

where

$$\begin{aligned} T_0 < T \leq T_s, C_0 &= \frac{2k_2}{C_{p1}}, C_1 = \frac{k_0 + k_1 T_0 + k_2 T_0^2}{C_{p0} + C_{p1} T_0} \\ C_2 &= \frac{-(k_2/2)T_0^2 + (k_1/2 - 2(C_{p0}/C_{p1})k_2)T_0}{C_{p0}/C_{p1} + T_0} \\ &+ \frac{-k_0/2 + (C_{p0}/C_{p1})k_1 - 2(C_{p0}/C_{p1})^2 k_2}{C_{p0}/C_{p1} + T_0} \end{aligned} \quad (2.3-9)$$

Integration of Eq. (2.3-8c) gives

$$x = \frac{1}{V\rho_0} (C_0(T_s - T) + C_1 \ln \frac{T_s - T_0}{T - T_0} + C_2 \frac{2}{C_{p1}} \ln \frac{|C_{p0} + (C_{p1}/2)(T_s + T_0)|}{|C_{p0} + (C_{p1}/2)(T + T_0)|}) \quad (2.3-10)$$

2.3.3. Case 2.3-3 (k : constant, linear or quadratic function of T , C_p : constant, ρ : constant)

Substituting Eq. (2.3-1) with $C_{p2} = 0$, $C_{p1} = 0$ and $C_{p0} \neq 0$ into Eq. (2.2-6) yields

$$x = \frac{1}{V\rho_0} \int_T^{T_s} \frac{k_0 + k_1 v + k_2 v^2}{C_{p0}(v - T_0)} dv \quad (2.3-11a)$$

$$= \frac{1}{V\rho_0} \int_T^{T_s} [\frac{k_1 + 2k_2 T_0}{C_{p0}} + \frac{k_0 + k_1 T_0 + k_2 T_0^2}{C_{p0}(v - T_0)} + \frac{k_2}{C_{p0}}(v - T_0)] dv \quad (2.3-11b)$$

where

$$T_0 < T \leq T_s \quad (2.3-11c)$$

Integration of Eq. (2.3-11b) gives

$$x = \frac{1}{V\rho_0} \left(-\frac{k_2}{2C_{p0}} T^2 - \frac{k_1 + k_2 T_0}{C_{p0}} T + \frac{k_2 T_s^2 + 2k_2 T_s T_0 + 2k_1 T_s}{2C_{p0}} + \frac{k_2 T_0^2 + k_1 T_0 + k_0}{C_{p0}} \ln \frac{|T_s - T_0|}{|T - T_0|} \right) \quad (2.3-12a)$$

When $k_2 = 0$, Eq. (2.3-12a) becomes

$$x = \frac{1}{V\rho_0} \left(\frac{k_1}{C_{p0}} (-T + T_s) + \frac{k_1 T_0 + k_0}{C_{p0}} \ln \frac{|T_s - T_0|}{|T - T_0|} \right) \quad (2.3-12b)$$

When $k_1 = k_2 = 0$ in Eq. (2.3-12a) we obtain

$$x = \frac{k_0}{V\rho_0 C_{p0}} \ln \frac{|T_s - T_0|}{|T - T_0|} \quad (2.3-12c)$$

Eq. (2.3-12c) is also represented by

$$T - T_0 = (T_s - T_0) \exp\left(-\frac{V}{\alpha} x\right) \quad (2.3-12d)$$

where $\alpha = k_0/(\rho_0 C_{p0})$. Eq. (2.3-12d) is the analytical solution [5,15,19] for the case of constant thermo-physical properties.

2.4. Calculation of the analytical solution for the single phase model

A calculated example of the analytical solution for the single phase model in the quasi-steady-state condition using Eq. (2.3-10) is shown in Fig. 2. The calculation was carried out with $T_s = 950$ K, $T_0 = 300$ K, $V = 7.5545 \times 10^{-5}$ m/s, $k = 0.12100 + 0.00048567T$ W/(m·K), $C_p = 644.35 + 1.4068T$ J/(kg·K), $\rho = 2310$ kg/m³, in which k , C_p and ρ are based on [21]. (In [21] two kinds of specific heats C_p ($C_p(s)$ ($T < 600$ K) and $C_p(l)$ ($T > 600$ K) are given; in this paper $C_p(l)$ is used for the whole temperature range of 300–950 K.) The surface recession rate V is calculated by using the equation of the relation [22,23] between T_s and V for the material of Teflon (In the calculation of V , material properties, etc. given in [23] were used). From the figure, it is seen that T is convex downward; as x increases T decreases. In the figure numerical results by the finite difference method (FDM) are also shown. The transient heat transfer problem of Eq. (2.1-8) with $ds/dt = V$ (a constant value) was solved by numerical calculation with the FDM. The initial condition is $T(x, 0) = T_0$, and the boundary conditions are $T(0, t) = T_s$ and $T(L_\infty, t) = T_0$, where L_∞ is set to be 0.05 m as a large value. T_s , T_0 , V , k , C_p and ρ are the same as those of the analytical solution. The time step for the calculation is 0.01 s. The calculated temperature for the time of 10000 s is adopted for the solution by the FDM. The figure shows that the

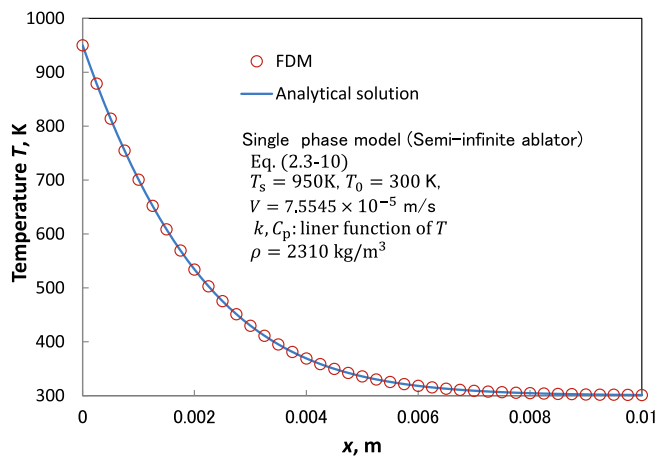


Fig. 2. Example of analytical solution for non-charring ablator (Quasi-steady single phase model).

analytical and numerical solutions agree well.

As for the validity of the analytical solution, each analytical solution in Section 2.3.1-2.3.3 was calculated and was compared with the solution by FDM. The analytical and numerical solutions agreed well.

Temperature distributions of the Teflon material by the single phase model in the quasi-steady state condition for $V = 7.5545 \times 10^{-5}$ m/s with various T_s (900, 950, 1000 K) and for $T_s = 950$ K with various V ($0.5 \times 7.5545 \times 10^{-5}$, 7.5545×10^{-5} , $2 \times 7.5545 \times 10^{-5}$ m/s) were calculated by using the analytical solution of Eq. (2.3-10). The calculation results are shown in Figs. 3 and 4. In this single phase model, the material is assumed to be a solid. In Fig. 3, it is seen that the temperature increases as T_s increases. In Fig. 4, it is seen that the temperature decreases as V increases. Within the parameter range of T_s and V in these figures, the effect of V upon the temperature distribution is larger than that of T_s .

3. Non-charring ablator composed of two phases

Various kinds of transient numerical simulation of the Teflon material with two phases using temperature-dependent thermo-physical properties have been carried out [10–14]. In the following, analytical solution for quasi-steady one-dimensional non-charring ablation in a semi-infinite material composed of two phases (the crystalline (solid) and the amorphous (gel) phases) such as Teflon with temperature-dependent thermo-physical properties is investigated. The crystalline (solid) phase region is located in the back of the amorphous (gel) phase region. Ablation is assumed to occur at the surface of the amorphous region. At the boundary between the amorphous and the crystalline regions (phase transition plane), the crystalline phase changes to the amorphous phase at the phase transition temperature. In the quasi-steady-state condition, the surface temperature is assumed to be constant and the surface is assumed to recede with a constant velocity. Thermo-physical properties (k , C_p , ρ) are assumed to be dependent on the temperature.

3.1. Ablation model and basic equations

Consider a one-dimensional semi-infinite solid made of a non-charring ablative material consisting of two phases (a two-phase

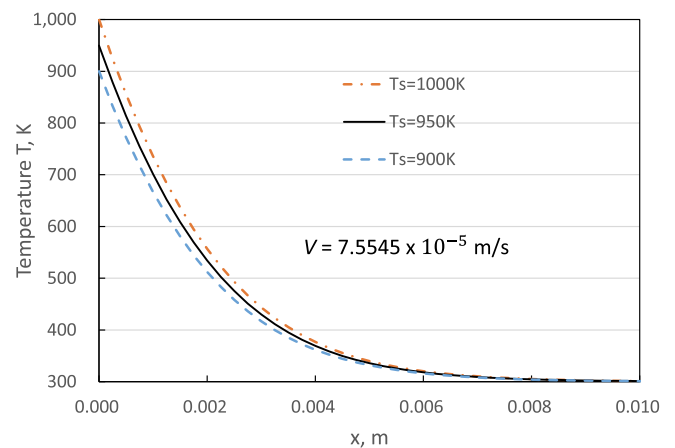


Fig. 3. Effect of T_s upon the temperature distribution (Quasi-steady single phase model).

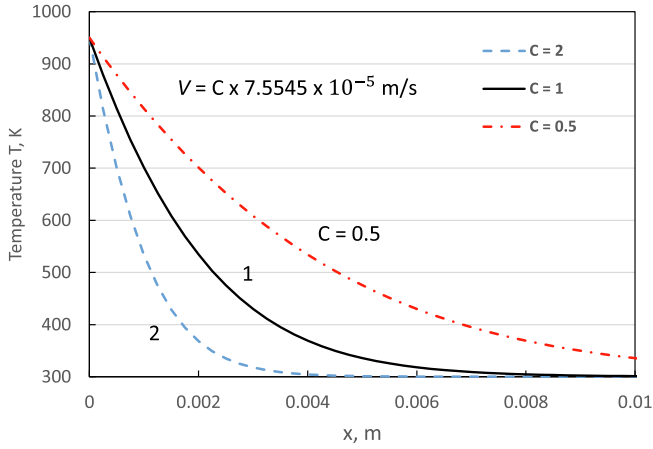


Fig. 4. Effect of V upon the temperature distribution (Quasi-steady single phase model).

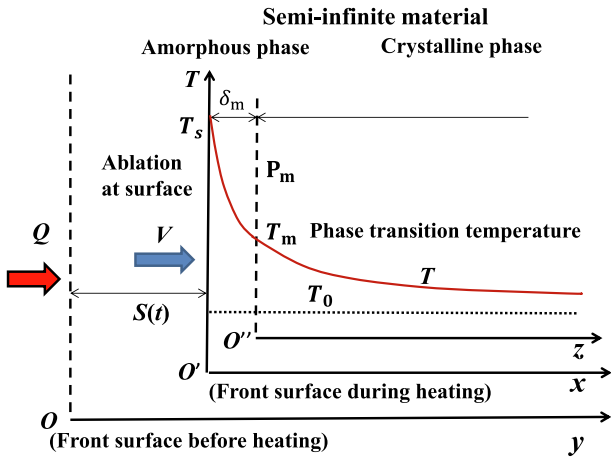


Fig. 5. Semi-infinite non-charring ablator (Quasi-steady two-phase model).

model) as shown in Fig. 5. The solid is assumed initially at a constant temperature T_0 , and then suddenly a constant heat flux Q is loaded on the front surface. If heating continues long enough with a high enough value of Q , the surface recedes by ablation at a constant velocity V and the surface keeps a constant value of the surface temperature T_s . In the quasi-steady-state condition, it is assumed that the material comprises two regions (the amorphous (gel) and the crystalline (solid) phase regions); ablation occurs at the surface of the amorphous phase region; at the phase transition plane P_m , the crystalline phase changes to the amorphous phase at a phase transition temperature T_m . The axes of x , y and z are considered: x represents the moving coordinate in which the origin O' is fixed at the moving surface, y represents the stationary coordinate in which the origin O is fixed at the initial plane of the surface, and z represents the moving coordinate in which the origin O'' is fixed at the phase transition plane P_m moving with the velocity of V_m . The length of OO'' is $S(t)$. The thermal conductivity k , the specific heat C_p and the density ρ of the material are assumed to be functions of temperature T . In this model, thermal expansion is neglected [10,13].

The one-dimensional unsteady heat conduction equation (the energy equation) for both phases in the semi-infinite material in the stationary coordinate system is given as

$$\rho C_p \frac{\partial T}{\partial t} \Big|_y = \frac{\partial}{\partial y} \left(k \frac{\partial T}{\partial y} \right) \Big|_t \quad (3.1-1a)$$

In the similar way to Section 2.1, in the moving coordinate system, the energy equation for both phases is expressed by

$$\rho C_p \frac{\partial T}{\partial t} \Big|_x = \frac{\partial}{\partial x} \left(k \frac{\partial T}{\partial x} \right) \Big|_t + \rho C_p \frac{dS}{dt} \frac{\partial T}{\partial x} \Big|_t \quad (3.1-1b)$$

(a) Crystalline phase region

In the quasi-steady-state condition the left side term of Eq. (3.1-1b) becomes zero, dS/dt becomes a constant V , and the temperature becomes the function of x only. Considering that z is represented by

$$z = x - \delta_m \quad (3.1-2)$$

where δ_m is the thickness of the amorphous region, we have

$$\frac{d}{dz} \left(k \frac{dT}{dz} \right) + \rho C_p V \frac{dT}{dz} = 0 \quad (3.1-3)$$

with the boundary conditions

$$T(0) = T_m \quad (3.1-4a)$$

$$T(z) \rightarrow T_0 \text{ as } z \rightarrow \infty \quad (3.1-4b)$$

From Eq. (3.1-4b) the following equation is also obtained.

$$dT(z)/dz = 0 \text{ as } z \rightarrow \infty \quad (3.1-4c)$$

In Eqs. (3.1-2) to (3.1-4c), δ_m , V , T_m and T_0 are constant, and k , C_p and ρ are functions of T .

(b) Amorphous phase region

In the similar way to the case of the crystalline phase region, the energy equation is given by

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + \rho C_p V \frac{dT}{dx} = 0 \quad (3.1-5)$$

The boundary conditions are

$$T(0) = T_s \quad (3.1-6a)$$

$$T(\delta_m) = T_m \quad (3.1-6b)$$

Energy balance at the phase transition plane gives [10,13]

$$-\left(k \frac{dT}{dx}\right)_{x=\delta_m-0} = -\left(k \frac{dT}{dx}\right)_{x=\delta_m+0} + V_m \rho_m h_m \quad (3.1-7)$$

where V_m is the velocity of the phase transition plane, ρ_m is $(\rho_a + \rho_c)/2$ (ρ_a, ρ_c : densities of the amorphous and the crystalline phases at the phase transition temperature T_m , respectively) and h_m is the latent heat of phase transition from the solid to the gel phase. In the quasi-steady-state condition, $V_m = V$. (In [14,23], the value of ρ_c instead of $(\rho_a + \rho_c)/2$ is used for ρ_m in Eq. (3.1-7).) In Eqs. (3.1-5) to (3.1-7), V , T_s , T_m , ρ_m , h_m and δ_m are constant, and k , C_p and ρ are functions of T .

As an example of application, the ablation model mentioned above can be applied to the situation in which the ablative material is under the heating test with constant Q . (If Q is constant, T_s and V are also constant in a quasi-steady state condition.)

3.2. Analytical solution for the two-phase model

(a) Crystalline phase region

Here, analytical solution for the boundary value problem of Eqs. (3.1-2) to (3.1-4c) is derived. Eq. (3.1-3) can be expressed as

$$\frac{d}{dz} \left(k \frac{dT}{dz} \right) + V \frac{d}{dz} \int_{T_0}^T \rho(u) C_p(u) du = 0 \quad (3.2-1)$$

Eq. (3.2-1) yields

$$\frac{d}{dz} \left[k \frac{dT}{dz} + V \int_{T_0}^T \rho(u) C_p(u) du \right] = 0 \quad (3.2-2)$$

Then

$$k \frac{dT}{dz} + V \int_{T_0}^T \rho(u) C_p(u) du = C \quad (3.2-3)$$

where C is a constant. Consideration of Eqs. (3.1-4b), (3.1-4c) and (3.2-3) gives $C = 0$, then we have

$$\frac{dT}{dz} = -\frac{V}{k} \int_{T_0}^T \rho(u) C_p(u) du \quad (3.2-4)$$

The reciprocal of Eq. (3.2-4) gives

$$\frac{dz}{dT} = -\frac{k(T)/V}{\int_{T_0}^T \rho(u) C_p(u) du} \quad (3.2-5)$$

In Eq. (3.2-5), T is defined in $T_0 < T \leq T_m$. Integrating Eq. (3.2-5) and using Eqs. (3.1-2) and (3.1-4a), we obtain

$$x = \delta_m + z \quad (3.2-6a)$$

$$z = \frac{1}{V} \int_T^{T_m} \frac{k(v)}{\int_{T_0}^v \rho(u) C_p(u) du} dv \quad (3.2-6b)$$

where δ_m is the thickness of the amorphous phase region given by Eq. (3.2-15). Apparently, the denominator in the integrand is positive for $T_0 < v \leq T_m$. Eqs. (3.2-6a) and (3.2-6b) express the analytical solution of the boundary value problem of Eqs. (3.1-2) to (3.1-4c) for the crystalline phase in the two-phase ablation model. It is noted that Eq. (3.2-6b) becomes Eq. (2.2-6), if T_m and z in Eq. (3.2-6b) are replaced by T_s and x , respectively, and ρ in Eq. (3.2-6b) becomes a constant.

In Section 3.3, simple analytical solutions in the solid region are derived for the case that $k(T)$, $C_p(T)$ and $\rho(T)$ are expressed by low degree polynomials of T .

(b) Amorphous phase region

Analytical solution for the boundary value problem of Eqs. (3.1-5) to (3.1-7) is derived below.

Eq. (3.1-5) can be expressed as

$$\frac{d}{dx} \left(k \frac{dT}{dx} \right) + V \frac{d}{dx} \int_{T_m}^T \rho(u) C_p(u) du = 0 \quad (3.2-7)$$

Then

$$\frac{d}{dx} \left[k \frac{dT}{dx} + V \int_{T_m}^T \rho(u) C_p(u) du \right] = 0 \quad (3.2-8)$$

Therefore

$$k \frac{dT}{dx} + V \int_{T_m}^T \rho(u) C_p(u) du = C \quad (3.2-9)$$

From Eqs. (3.1-6b), (3.1-7), (3.2-4), (3.2-6a) and (3.2-9), we obtain

$$C = -V \int_{T_0}^{T_m} \rho(u) C_p(u) du - V \rho_m h_m \quad (3.2-10)$$

Substitution of Eq. (3.2-10) into Eq. (3.2-9) yields

$$\frac{dT}{dx} = \frac{-V \int_{T_m}^T \rho(u) C_p(u) du - V \int_{T_0}^{T_m} \rho(u) C_p(u) du - V \rho_m h_m}{k(T)} \quad (3.2-11)$$

The reciprocal of Eq. (3.2-11) gives

$$\frac{dx}{dT} = -\frac{k(T)}{V} \frac{1}{\int_{T_m}^T \rho(u) C_p(u) du + \int_{T_0}^{T_m} \rho(u) C_p(u) du + \rho_m h_m} \quad (3.2-12)$$

In Eq. (3.2-12), T is defined in $T_m \leq T \leq T_s$. Eqs. (3.1-6a) and (3.2-

12) yield the solution

$$x = \frac{1}{V} \int_T^{T_s} \frac{k(v)}{\int_{T_m}^v \rho(u) C_p(u) du + \varphi_1 + \varphi_2} dv \quad (3.2-13)$$

where φ_1 and φ_2 are given by

$$\varphi_1 = \int_{T_0}^{T_m} \rho(u) C_p(u) du, \varphi_2 = \rho_m h_m \quad (3.2-14)$$

When ρ and C_p are expressed by polynomials of T , φ_1 can be easily calculated. Apparently, the denominator in the integrand in Eq. (3.2-13) is positive for $T_m \leq v \leq T_s$. From Eqs. (3.1-6b) and (3.2-13), we obtain

$$\delta_m = \frac{1}{V} \int_{T_m}^{T_s} \frac{k(v)}{\int_{T_m}^v \rho(u) C_p(u) du + \varphi_1 + \varphi_2} dv \quad (3.2-15)$$

Eqs. (3.2-13) to (3.2-15) express the analytical solution for the boundary value problem of Eqs. (3.1-5) to (3.1-7) for the amorphous region.

In Section 3.4, simple analytical solutions in the amorphous region are derived for the case that $k(T)$, $C_p(T)$ and $\rho(T)$ are expressed by low degree polynomials of T .

The ablation model in this section can be applied to the Teflon material (two-phase model which consists of crystalline and amorphous phases). At temperatures below the phase transition temperature T_m (600 K), Teflon is in the crystalline phase. A major phase transition occurs at this temperature and the melted Teflon is amorphous. The phase change requires a heat of transition of h_m and the change in physical properties occur. As the temperature of Teflon is raised well above T_m , depolymerization occurs.

3.3. Simple analytical solution for the solid region in the two-phase model

It is assumed that k , C_p and ρ are expressed by the following equations

$$\begin{aligned} k(T) &= k_0 + k_1 T + k_2 T^2, \\ C_p(T) &= C_{p0} + C_{p1} T + C_{p2} T^2, \\ \rho(T) &= \rho_0 + \rho_1 T + \rho_2 T^2 \end{aligned} \quad (3.3-1)$$

where $k(T)$, $C_p(T)$, and $\rho(T)$ are positive, and the case of both $C_{p2} \neq 0$ and $\rho_2 \neq 0$ is excluded. The degree N of the polynomial for $C_p(T) \times \rho(T)$ is limited to $N \leq 3$.

3.3.1. Case 3.3-1 (k : constant, linear or quadratic function of T , C_p : Linear function of T , ρ : Quadratic function of T)

Substituting Eq. (3.3-1) with $C_{p2} = 0$, $C_{p1} \neq 0$ and $\rho_2 \neq 0$ into Eq. (3.2-6b) yields

$$z = \frac{1}{V} \int_T^{T_m} \frac{k_0 + k_1 v + k_2 v^2}{\int_{T_0}^v (\rho_0 + \rho_1 u + \rho_2 u^2)(C_{p0} + C_{p1} u) du} dv \quad (3.3-2a)$$

$$= \frac{1}{V} \int_T^{T_m} \left[\frac{e}{v - T_0} + \frac{fv^2 + gv + h}{a_3 v^3 + a_2 v^2 + a_1 v + a_0} \right] dv \quad (3.3-2b)$$

where

$$\begin{aligned} e &= \frac{k_2 T_0^2 + k_1 T_0 + k_0}{a_3 T_0^3 + a_2 T_0^2 + a_1 T_0 + a_0}, f = -a_3 e, g = k_2 - e(a_2 + a_3 T_0), h = \frac{a_0 e - k_0}{T_0}, \\ a_3 &= \frac{C_{p1} \rho_2}{4}, a_2 = \frac{C_{p0} \rho_2 + C_{p1} \rho_1}{3} + \frac{C_{p1} \rho_2 T_0}{4}, \\ a_1 &= \frac{C_{p0} \rho_1 + C_{p1} \rho_0}{2} + \frac{C_{p0} \rho_2 + C_{p1} \rho_1}{3} T_0 + \frac{C_{p1} \rho_2 T_0^2}{4}, \\ a_0 &= C_{p0} \rho_0 + \frac{C_{p0} \rho_1 + C_{p1} \rho_0 T_0}{2} + \frac{C_{p0} \rho_2 + C_{p1} \rho_1 T_0^2}{3} + \frac{C_{p1} \rho_2 T_0^3}{4} \end{aligned} \quad (3.3-2c)$$

Then Eq. (3.3-2b) is expressed by

$$z = \frac{1}{V} (e \ln \frac{|T_m - T_0|}{|T - T_0|} + p \ln \frac{|T_m - \alpha|}{|T - \alpha|} + q I_B(T_m, T) + r I_C(T_m, T)) \quad (3.3-3)$$

where

$$\begin{aligned} T_0 < T \leq T_m, p &= \frac{f\alpha^2 + g\alpha + h}{A\alpha^2 + B\alpha + C}, q = \frac{Cg - Bh + (Cf - Ah)\alpha}{A\alpha^2 + B\alpha + C}, \\ r &= \frac{Cf - Ah + (Bf - Ag)\alpha}{A\alpha^2 + B\alpha + C}, A = a_3, B = a_2 + \alpha a_3, C = -a_0/\alpha, \\ I_B(T_m, T) &= \int_T^{T_m} \frac{1}{Av^2 + Bv + C} dv, \quad I_C(T_m, T) = \int_T^{T_m} \frac{v}{Av^2 + Bv + C} dv \end{aligned} \quad (3.3-4)$$

In Eq. (3.3-4), $I_B(T_m, T)$ and $I_C(T_m, T)$ can be calculated by Eqs. (2.3-5) to (2.3-7), in which T_s is replaced by T_m with A, B and C given by Eq. (3.3-4). α is a real number and one of the three kinds of roots for

$$a_3 v^3 + a_2 v^2 + a_1 v + a_0 = 0 \quad (3.3-5)$$

which is given as follows [24].

Dividing both sides of Eq. (3.3-5) by a_3 yields

$$v^3 + A_2 v^2 + A_1 v + A_0 = 0 \quad (3.3-6)$$

Let

$$v = X - A_2/3 \quad (3.3-7)$$

Then Eq. (3.3-6) becomes

$$X^3 + PX + Q = 0 \quad (3.3-8a)$$

where

$$P = A_1 - A_2^2/3, Q = A_0 - A_1 A_2/3 + (2/27) A_2^3 \quad (3.3-8b)$$

One root for Eqs. (3.3-8a) and (3.3-8b) is

$$X = W - P/(3W) \quad (3.3-9)$$

where

$$W = (-Q/2 + \sqrt{DD})^{1/3}, DD = (Q/2)^2 + (P/3)^3 \quad (3.3-10)$$

Then α which is one root of Eq. (3.3-6) is given by

$$\alpha = W - P/(3W) - A_2/3 \quad (3.3-11)$$

Note that A_0, A_1, A_2, P and Q are real numbers. According to the value of DD , α which is a real number is given by (a) or (b).

(a) When $DD \geq 0$, we regard W as a real cubic root of $-Q/2 + \sqrt{DD}$, then α which is given by Eq. (3.3-11) is apparently a real number.

(b) When $DD < 0$, from Eq. (3.3-10) we have $(P/3)^3 < -(Q/2)^2$, which shows that P is negative. Eq. (3.3-10) is expressed by

$$W = (-Q/2 + \sqrt{-DD} i)^{1/3} \quad (3.3-12a)$$

By using the polar form, Eq. (3.3-12a) can be expressed by

$$W = \{R(\cos\theta + i\sin\theta)\}^{1/3} \quad (3.3-12b)$$

where

$$\begin{aligned} R &= (-P/3)^{3/2}, \\ \cos\theta &= \frac{-Q/2}{R}, \end{aligned} \quad (3.3-12c)$$

$$\sin\theta = \frac{\sqrt{-DD}}{R}, 0 < \theta < \pi$$

Then

$$W = R^{1/3}(\cos(\theta/3) + i\sin(\theta/3)) \quad (3.3-12d)$$

Substituting Eq. (3.3-12d) into Eq. (3.3-11), we have α which is a real

number as shown below

$$\alpha = 2R^{1/3}\cos(\theta/3) - A_2/3 \quad (3.3-13a)$$

$$= 2\sqrt{(-P/3)}\cos(\theta/3) - A_2/3 \quad (3.3-13b)$$

where

$$\theta = \cos^{-1} \frac{-Q/2}{R} = \cos^{-1} \frac{-Q/2}{(-P/3)^{3/2}}, 0 < \theta < \pi \quad (3.3-13c)$$

3.3.2. Case 3.3-2 (k : constant, linear or quadratic function of T , C_p : Quadratic function of T , ρ : Linear function of T)

Substituting Eq. (3.3-1) with $C_{p2} \neq 0$, and $\rho_2 = 0$, $\rho_1 \neq 0$ into Eq. (3.2-6b) yields

$$z = \frac{1}{V} \int_T^{T_m} \frac{k_0 + k_1 v + k_2 v^2}{\int_{T_0}^v (\rho_0 + \rho_1 u)(C_{p0} + C_{p1} u + C_{p2} u^2) du} dv \quad (3.3-14a)$$

$$= \frac{1}{V} \int_T^{T_m} \left[\frac{e}{v - T_0} + \frac{fv^2 + gv + h}{a_3 v^3 + a_2 v^2 + a_1 v + a_0} \right] dv \quad (3.3-14b)$$

where

$$\begin{aligned} T_0 < T \leq T_m, e &= \frac{k_2 T_0^2 + k_1 T_0 + k_0}{a_3 T_0^3 + a_2 T_0^2 + a_1 T_0 + a_0}, f = -a_3 e, g = k_2 - e(a_2 + a_3 T_0) \\ h &= \frac{a_0 e - k_0}{T_0}, a_3 = \frac{C_{p2} \rho_1}{4}, a_2 = \frac{C_{p2} \rho_0 + C_{p1} \rho_1}{3} + \frac{C_{p2} \rho_1 T_0}{4}, \\ a_1 &= \frac{C_{p1} \rho_0 + C_{p0} \rho_1}{2} + \frac{C_{p2} \rho_0 + C_{p1} \rho_1}{3} T_0 + \frac{C_{p2} \rho_1 T_0^2}{4}, \\ a_0 &= C_{p0} \rho_0 + \frac{C_{p1} \rho_0 + C_{p0} \rho_1}{2} T_0 + \frac{C_{p2} \rho_0 + C_{p1} \rho_1}{3} T_0^2 + \frac{C_{p2} \rho_1 T_0^3}{4} \end{aligned} \quad (3.3-14c)$$

Then

$$z = \frac{1}{V} (e \ln \frac{|T_m - T_0|}{|T - T_0|} + p \ln \frac{|T_m - \alpha|}{|T - \alpha|} + q I_B(T_m, T) + r I_C(T_m, T)) \quad (3.3-15)$$

where

$$\begin{aligned} T_0 < T \leq T_m, p &= \frac{f\alpha^2 + g\alpha + h}{A\alpha^2 + B\alpha + C}, q = \frac{Cg - Bh + (Cf - Ah)\alpha}{A\alpha^2 + B\alpha + C}, \\ r &= \frac{Cf - Ah + (Bf - Ag)\alpha}{A\alpha^2 + B\alpha + C}, A = a_3, B = a_2 + \alpha a_3, C = -a_0/\alpha \end{aligned} \quad (3.3-16a)$$

$$I_B(T_m, T) = \int_T^{T_m} \frac{1}{Av^2 + Bv + C} dv,$$

$$I_C(T_m, T) = \int_T^{T_m} \frac{v}{Av^2 + Bv + C} dv$$

$I_B(T_m, T)$ and $I_C(T_m, T)$ can be calculated by using Eqs. (2.3-5) to (2.3-7), in which T_s is replaced by T_m with A, B and C given by Eq. (3.3-16a). α is a real number and one of the three kinds of roots for

$$a_3 v^3 + a_2 v^2 + a_1 v + a_0 = 0 \quad (3.3-16b)$$

which is given in the same way as in Section 3.3.1.

3.3.3. Case 3.3-3 (k : constant, linear or quadratic function of T , C_p : constant, ρ : Quadratic function of T)

Substituting Eq. (3.3-1) with $C_{p0} \neq 0$, $C_{p1} = C_{p2} = 0$ and $\rho_2 \neq 0$ into Eq. (3.2-6b) yields

$$z = \frac{1}{V} \int_T^{T_m} \frac{k_0 + k_1 v + k_2 v^2}{\int_{T_0}^v (\rho_0 + \rho_1 u + \rho_2 u^2) C_{p0} du} dv \quad (3.3-17a)$$

$$= \frac{1}{V} \int_T^{T_m} \left[\frac{p}{v - T_0} + \frac{q + rv}{Av^2 + Bv + C} \right] dv \quad (3.3-17b)$$

Then

$$z = \frac{1}{V} (p \ln \frac{|T_m - T_0|}{|T - T_0|} + q I_B(T_m, T) + r I_C(T_m, T)) \quad (3.3-17c)$$

where

$$\begin{aligned} T_0 < T \leq T_m, p &= \frac{k_2 T_0^2 + k_1 T_0 + k_0}{A T_0^2 + B T_0 + C}, \\ q &= \frac{(C k_2 - A k_0) T_0 + C k_1 - B k_0}{A T_0^2 + B T_0 + C}, r = \frac{(B k_2 - A k_1) T_0 + C k_2 - A k_0}{A T_0^2 + B T_0 + C}, \\ A &= \frac{\rho_2}{3} C_{p0}, B = (\frac{\rho_1}{2} + \frac{\rho_2}{3} T_0) C_{p0}, C = (\rho_0 + \frac{\rho_1}{2} T_0 + \frac{\rho_2}{3} T_0^2) C_{p0} \end{aligned} \quad (3.3-17d)$$

$I_B(T_m, T)$ and $I_C(T_m, T)$ can be calculated by using Eqs. (2.3-5) to (2.3-7), in which T_s is replaced by T_m with A, B and C given by Eq. (3.3-17d).

3.3.4. Case 3.3-4 (k : constant, linear or quadratic function of T , C_p : Linear function of T , ρ : Linear function of T)

Substituting Eq. (3.3-1) with $C_{p2} = 0, C_{p1} \neq 0$, and $\rho_2 = 0, \rho_1 \neq 0$ into Eq. (3.2-6b) yields

$$z = \frac{1}{V} \int_{T_0}^{T_m} \frac{k_0 + k_1 v + k_2 v^2}{\int_{T_0}^v (\rho_0 + \rho_1 u) (C_{p0} + C_{p1} u) du} dv \quad (3.3-18a)$$

$$= \frac{1}{V} \int_T^{T_m} [\frac{p}{v - T_0} + \frac{q + rv}{A v^2 + B v + C}] dv \quad (3.3-18b)$$

Then

$$z = \frac{1}{V} (p \ln \frac{|T_m - T_0|}{|T - T_0|} + q I_B(T_m, T) + r I_C(T_m, T)) \quad (3.3-19a)$$

where

$$\begin{aligned} T_0 < T \leq T_m, p &= \frac{k_2 T_0^2 + k_1 T_0 + k_0}{A T_0^2 + B T_0 + C}, \\ q &= \frac{(C k_2 - A k_0) T_0 + C k_1 - B k_0}{A T_0^2 + B T_0 + C}, r = \frac{(B k_2 - A k_1) T_0 + C k_2 - A k_0}{A T_0^2 + B T_0 + C}, \\ A &= \rho_1 C_{p1}/3, B = (C_{p0} \rho_1 + C_{p1} \rho_0)/2 + (1/3) C_{p1} \rho_1 T_0, \\ C &= C_{p0} \rho_0 + (1/2)(C_{p0} \rho_1 + C_{p1} \rho_0) T_0 + (1/3) C_{p1} \rho_1 T_0^2 \end{aligned} \quad (3.3-19b)$$

$I_B(T_m, T)$ and $I_C(T_m, T)$ can be calculated by using Eqs. (2.3-5) to (2.3-7), in which T_s is replaced by T_m with A, B and C given by Eq. (3.3-19b).

3.3.5. Case 3.3-5 (k : constant, linear or quadratic function of T , C_p : Quadratic function of T , ρ : constant)

Substituting Eq. (3.3-1) with $C_{p2} \neq 0$, and $\rho_2 = \rho_1 = 0, \rho_0 \neq 0$ into Eq. (3.2-6b) yields the solution. This case becomes the same as Case 2.3-1, if T_s and x are replaced by T_m and z , respectively. Then Eqs. (2.3-2a) to (2.3-7) give the solution of Case 3.3-5 with T_s and x replaced by T_m and

$$\begin{aligned} T_m \leq T \leq T_s, a_0 &= -(1/3) C_{p1} \rho_1 T_m^3 - (1/2)(C_{p0} \rho_1 + C_{p1} \rho_0) T_m^2 - C_{p0} \rho_0 T_m + \varphi_1 + \varphi_2, \\ \varphi_1, \varphi_2 : \text{Eq. (3.2-14)}, a_1 &= C_{p0} \rho_0, a_2 = (C_{p0} \rho_1 + C_{p1} \rho_0)/2, a_3 = C_{p1} \rho_1/3 \end{aligned} \quad (3.4-2b)$$

z , respectively.

3.3.6. Case 3.3-6 (k : constant, linear or quadratic function of T , C_p : constant, ρ : Linear function of T)

Substituting Eq. (3.3-1) with $C_{p1} = C_{p2} = 0, C_{p0} \neq 0$, and $\rho_2 = 0, \rho_1 \neq 0$ into Eq. (3.2-6b) yields

$$z = \frac{1}{V C_{p0}} (C_0 (T_m - T) + C_1 \ln \frac{|T_m - T_0|}{|T - T_0|} + C_2 \frac{2}{\rho_1} \ln \frac{|\rho_0 + (\rho_1/2)(T_m + T_0)|}{|\rho_0 + (\rho_1/2)(T + T_0)|}) \quad (3.3-20a)$$

where

$$\begin{aligned} T_0 < T \leq T_m, C_0 &= \frac{2k_2}{\rho_1}, C_1 = \frac{k_0 + k_1 T_0 + k_2 T_0^2}{\rho_0 + \rho_1 T_0} \\ C_2 &= \frac{-(k_2/2) T_0^2 + (k_1/2 - 2(\rho_0/\rho_1) k_2) T_0}{\rho_0/\rho_1 + T_0} + \frac{-k_0/2 + (\rho_0/\rho_1) k_1 - 2(\rho_0/\rho_1)^2 k_2}{\rho_0/\rho_1 + T_0} \end{aligned} \quad (3.3-20b)$$

3.3.7. Case 3.3-7 (k : constant, linear or quadratic function of T , C_p : Linear function of T , ρ : constant)

Substituting Eq. (3.3-1) with $C_{p2} = 0, C_{p1} \neq 0$, and $\rho_2 = \rho_1 = 0, \rho_0 \neq 0$ into Eq. (3.2-6b) yields the solution. This case becomes the same as Case 2.3-2, if T_s and x are replaced by T_m and z , respectively. Then Eqs. (2.3-8a) to (2.3-10) give the solution of Case 3.3-7 with T_s and x replaced by T_m and z , respectively.

3.3.8. Case 3.3-8 (k : constant, linear or quadratic function of T , C_p : constant, ρ : constant)

Substituting Eq. (3.3-1) with $C_{p2} = C_{p1} = 0, C_{p0} \neq 0$, and $\rho_2 = \rho_1 = 0, \rho_0 \neq 0$ into Eq. (3.2-6b) yields the solution. This case becomes the same as Case 2.3-3, if T_s and x are replaced by T_m and z , respectively. Then Eqs. (2.3-11a) to (2.3-12d) give the solution of Case 3.3-8 with T_s and x replaced by T_m and z , respectively.

3.4. Simple analytical solution for the amorphous region in the two-phase model

It is assumed that k, C_p and ρ are expressed by the following equations

$$\begin{aligned} k(T) &= k_0 + k_1 T + k_2 T^2, \\ C_p(T) &= C_{p0} + C_{p1} T + C_{p2} T^2, \\ \rho(T) &= \rho_0 + \rho_1 T + \rho_2 T^2 \end{aligned} \quad (3.4-1)$$

where $k(T), C_p(T)$ and $\rho(T)$ are positive. The degree N of the polynomial for $C_p(T) \times \rho(T)$ is limited to two or less.

3.4.1. Case 3.4-1 (k : constant, linear or quadratic function of T , C_p : Linear function of T , ρ : Linear function of T)

Substitution of Eq. (3.4-1) with $C_{p2} = 0, C_{p1} \neq 0$, and $\rho_2 = 0, \rho_1 \neq 0$ into Eq. (3.2-13) yields

$$x = \frac{1}{V} \int_T^{T_s} \frac{k_2 v^2 + k_1 v + k_0}{a_3 v^3 + a_2 v^2 + a_1 v + a_0} dv \quad (3.4-2a)$$

where

Eq. (3.4-2a) can be expressed by

$$x = \frac{1}{V} \int_T^{T_s} \frac{k_2 v^2 + k_1 v + k_0}{(v - \alpha)(A v^2 + B v + C)} dv \quad (3.4-3a)$$

$$= \frac{1}{V} \int_T^{T_s} [\frac{p}{v - \alpha} + \frac{q + rv}{A v^2 + B v + C}] dv \quad (3.4-3b)$$

where

$$p = \frac{k_2\alpha^2 + k_1\alpha + k_0}{A\alpha^2 + B\alpha + C}, q = \frac{Ck_1 - Bk_0 + (Ck_2 - Ak_0)\alpha}{A\alpha^2 + B\alpha + C},$$

$$r = \frac{Ck_2 - Ak_0 + (Bk_2 - Ak_1)\alpha}{A\alpha^2 + B\alpha + C}, A = a_3, B = a_2 + aa_3, C = -a_0/\alpha$$
(3.4-3c)

α is a real number and one of the three kinds of roots for

$$a_3v^3 + a_2v^2 + a_1v + a_0 = 0$$
(3.4-4)

which is obtained in the same way as that of Case 3.3-1. Then we have the solution

$$x = \frac{1}{V} (p \ln \frac{|T_s - \alpha|}{|T - \alpha|} + q I_B(T_s, T) + r I_C(T_s, T))$$
(3.4-5)

where $I_B(T_s, T)$ and $I_C(T_s, T)$ can be calculated by using Eqs. (2.3-5) to (2.3-7) with A, B and C given by Eq. (3.4-3c). In each of Eqs. (3.4-2a), (3.4-3a), (3.4-3b) and Eq. (3.4-5), if T is replaced by T_m , x becomes δ_m .

3.4.2. Case 3.4-2 (k : constant, linear or quadratic function of T , C_p : Quadratic function of T , ρ : constant)

Substitution of Eq. (3.4-1) with $C_{p2} \neq 0, \rho_2 = \rho_1 = 0, \rho_0 \neq 0$ into Eq. (3.2-13) yields

$$x = \frac{1}{V} \int_T^{T_s} \frac{k_2v^2 + k_1v + k_0}{a_3v^3 + a_2v^2 + a_1v + a_0} dv$$
(3.4-6a)

where

$$T_m \leq T \leq T_s, a_0 = -(1/3)C_{p2}\rho_0T_m^3 - (1/2)C_{p1}\rho_0T_m^2 - C_{p0}\rho_0T_m + \varphi_1 + \varphi_2,$$

$$\varphi_1, \varphi_2 : \text{Eq. (3.2-14)}, a_1 = C_{p0}\rho_0, a_2 = C_{p1}\rho_0/2, a_3 = C_{p2}\rho_0/3$$
(3.4-6b)

From Eq. (3.4-6a)

$$x = \frac{1}{V} \int_T^{T_s} \left[\frac{p}{v - \alpha} + \frac{q + rv}{Av^2 + Bv + C} \right] dv$$
(3.4-6c)

Then we have

$$x = \frac{1}{V} (p \ln \frac{|T_s - \alpha|}{|T - \alpha|} + q I_B(T_s, T) + r I_C(T_s, T))$$
(3.4-7a)

where

$$p = \frac{k_2\alpha^2 + k_1\alpha + k_0}{A\alpha^2 + B\alpha + C}, q = \frac{Ck_1 - Bk_0 + (Ck_2 - Ak_0)\alpha}{A\alpha^2 + B\alpha + C},$$

$$r = \frac{Ck_2 - Ak_0 + (Bk_2 - Ak_1)\alpha}{A\alpha^2 + B\alpha + C}, A = a_3, B = a_2 + aa_3, C = -a_0/\alpha$$
(3.4-7b)

$I_B(T_s, T)$ and $I_C(T_s, T)$ can be calculated by using Eqs. (2.3-5) to (2.3-

$$T_m \leq T \leq T_s, A = \rho_0 C_{p1}/2, B = \rho_0 C_{p0}, C = -(\rho_0 C_{p1}/2)T_m^2 - \rho_0 C_{p0}T_m + \varphi_1 + \varphi_2,$$

$$\varphi_1, \varphi_2 : \text{Eq. (3.2-14)}, p = k_2/A, q = k_0 - k_2C/A, r = k_1 - k_2B/A$$
(3.4-11b)

7) with A, B and C given by Eq. (3.4-7b). α is a real number and one of the three kinds of roots for Eq. (3.4-4) with a_0 to a_3 given by Eq. (3.4-6b). In each of Eqs. (3.4-6a), (3.4-6c) and (3.4-7a), if T is replaced by T_m , x becomes δ_m .

3.4.3. Case 3.4-3 (k : constant, linear or quadratic function of T , C_p : constant, ρ : Quadratic function of T)

Substitution of Eq. (3.4-1) with $C_{p2} = C_{p1} = 0, C_{p0} \neq 0$, and $\rho_2 \neq 0$ into Eq. (3.2-13) yields

$$x = \frac{1}{V} \int_T^{T_s} \frac{k_2v^2 + k_1v + k_0}{a_3v^3 + a_2v^2 + a_1v + a_0} dv$$
(3.4-8a)

where

$$T_m \leq T \leq T_s, a_0 = -(1/3)C_{p0}\rho_2T_m^3 - (1/2)C_{p0}\rho_1T_m^2 - C_{p0}\rho_0T_m + \varphi_1 + \varphi_2,$$

$$\varphi_1, \varphi_2 : \text{Eq. (3.2-14)}, a_1 = C_{p0}\rho_0, a_2 = C_{p0}\rho_1/2, a_3 = C_{p0}\rho_2/3$$
(3.4-8b)

From Eq. (3.4-8a)

$$x = \frac{1}{V} \int_T^{T_s} \left[\frac{p}{v - \alpha} + \frac{q + rv}{Av^2 + Bv + C} \right] dv$$
(3.4-8c)

Then we have

$$x = \frac{1}{V} (p \ln \frac{|T_s - \alpha|}{|T - \alpha|} + q I_B(T_s, T) + r I_C(T_s, T))$$
(3.4-9a)

where

$$p = \frac{k_2\alpha^2 + k_1\alpha + k_0}{A\alpha^2 + B\alpha + C}, q = \frac{Ck_1 - Bk_0 + (Ck_2 - Ak_0)\alpha}{A\alpha^2 + B\alpha + C},$$

$$r = \frac{Ck_2 - Ak_0 + (Bk_2 - Ak_1)\alpha}{A\alpha^2 + B\alpha + C}, A = a_3, B = a_2 + aa_3, C = -a_0/\alpha$$
(3.4-9b)

$I_B(T_s, T)$ and $I_C(T_s, T)$ can be calculated by using Eqs. (2.3-5) to (2.3-7) with A, B and C given by Eq. (3.4-9b). α is a real number and one of the three kinds of roots for Eq. (3.4-4) with a_0 to a_3 given by Eq. (3.4-8b). In Eqs. (3.4-8a), (3.4-8c) and (3.4-9a), if T is replaced by T_m , x becomes δ_m .

3.4.4. Case 3.4-4 (k : constant, linear or quadratic function of T , C_p : Linear function of T , ρ : constant)

Substituting Eq. (3.4-1) with $C_{p2} = 0, C_{p1} \neq 0$, and $\rho_2 = \rho_1 = 0, \rho_0 \neq 0$ into Eq. (3.2-13), we obtain

$$x = \frac{1}{V} \int_T^{T_s} \frac{k_2v^2 + k_1v + k_0}{Av^2 + Bv + C} dv$$
(3.4-10a)

$$= \frac{1}{V} \int_T^{T_s} \left[p + \frac{q + rv}{Av^2 + Bv + C} \right] dv$$
(3.4-10b)

Then

$$x = \frac{1}{V} (p(T_s - T) + q I_B(T_s, T) + r I_C(T_s, T))$$
(3.4-11a)

where

$I_B(T_s, T)$ and $I_C(T_s, T)$ can be calculated by using Eqs. (2.3-5) to (2.3-7) with A, B and C given by Eq. (3.4-11b). In each of Eqs. (3.4-10a), (3.4-10b) and (3.4-11a), if T is replaced by T_m , x becomes δ_m .

3.4.5. Case 3.4-5 (k : constant, linear or quadratic function of T , C_p : constant, ρ : Linear function of T)

Substituting Eq. (3.4-1) with $C_{p2} = C_{p1} = 0$, $C_{p0} \neq 0$, and $\rho_2 = 0$, $\rho_1 \neq 0$ into Eq. (3.2-13), we obtain

$$x = \frac{1}{V} \int_T^{T_s} \frac{k_2 v^2 + k_1 v + k_0}{A v^2 + B v + C} dv \quad (3.4-12a)$$

$$= \frac{1}{V} \int_T^{T_s} [p + \frac{q + rv}{A v^2 + B v + C}] dv \quad (3.4-12b)$$

Then we have

$$x = \frac{1}{V} (p(T_s - T) + qI_B(T_s, T) + rI_C(T_s, T)) \quad (3.4-13a)$$

where

$$T_m \leq T \leq T_s, A = \rho_1 C_{p0}/2, B = \rho_0 C_{p0}, C = -(\rho_1 C_{p0}/2)T_m^2 - \rho_0 C_{p0}T_m + \varphi_1 + \varphi_2, \\ \varphi_1, \varphi_2: \text{Eq. (3.2-14)}, p = k_2/A, q = k_0 - k_2 C/A, r = k_1 - k_2 B/A \quad (3.4-13b)$$

$I_B(T_s, T)$ and $I_C(T_s, T)$ can be calculated by using Eqs. (2.3-5) to (2.3-7) with A, B and C given by Eq. (3.4-13b). In each of Eqs. (3.4-12a), (3.4-12b) and (3.4-13a), if T is replaced by T_m , x becomes δ_m .

3.4.6. Case 3.4-6 (k : constant, linear or quadratic function of T , C_p : constant, ρ : constant)

Substituting Eq. (3.4-1) with $C_{p2} = C_{p1} = 0$, $C_{p0} \neq 0$, and $\rho_2 = \rho_1 = 0$, $\rho_0 \neq 0$ into Eq. (3.2-13), we obtain

$$x = \frac{1}{V} \left(\frac{p(T_s^2 - T^2)}{2} + q(T_s - T) + \frac{r}{\rho_0 C_{p0}} \ln \frac{|T_s + \varphi/(\rho_0 C_{p0})|}{|T + \varphi/(\rho_0 C_{p0})|} \right) \quad (3.4-14a)$$

where

$$T_m \leq T \leq T_s, \varphi = \varphi_1 + \varphi_2 - \rho_0 C_{p0} T_m, \varphi_1, \varphi_2: \text{Eq. (3.2-14)}, p = k_2/(\rho_0 C_{p0}) \\ q = (k_1 \rho_0 C_{p0} - \varphi k_2)/(\rho_0 C_{p0})^2, \\ r = (k_0(\rho_0 C_{p0})^2 - k_1 \rho_0 C_{p0} \varphi + k_2 \varphi^2)/(\rho_0 C_{p0})^2 \quad (3.4-14b)$$

In Eq. (3.4-14a), if T is replaced by T_m , x becomes δ_m .

3.5. Calculation of the analytical solution for the two-phase model

A calculated example of the analytical solution for the two-phase model in the quasi-steady-state condition by using Eqs. (3.4-5) and (3.3-3) for the amorphous and the crystalline regions, respectively, is shown in Fig. 6. The calculation conditions in the amorphous region are

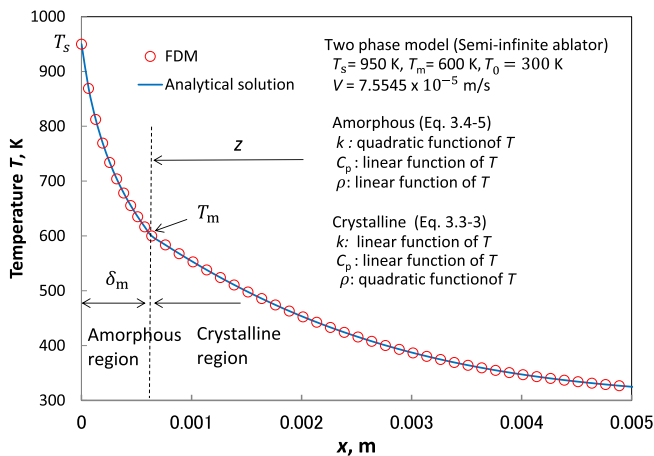


Fig. 6. Example of analytical solution for non-charring ablator (Quasi-steady two-phase model).

$T_s = 950$ K, $T_m = 600$ K, $V = 7.5545 \times 10^{-5}$ m/s, $k = 0.88090 - 0.0013984T + 5.8197 \times 10^{-7}T^2$ W/(mK), $C_p = 904.35 + 0.65314T$ J/(kg-K), $\rho = 2070 - 0.7T$ kg/m³ and $h_m = 58615$ J/kg. The calculation conditions in the crystalline region are $T_m = 600$ K, $T_0 = 300$ K, $V = 7.5545 \times 10^{-5}$ m/s, $k = 0.050242 + 0.00061420T$ W/(m-K), $C_p = 514.98 + 1.5629T$ J/(kg-K), $\rho = 2119 + 0.792T - 2.105 \times 10^{-3}T^2$ kg/m³. In both regions, T_m, k, C_p and ρ are based on [11]. The surface recession rate V is calculated by using the equation of the relation [22,23] between T_s and V for the material of Teflon (In the calculation of V , material properties, etc. given in [23] were used).

From the figure, it is seen that for both regions the analytical solution of T is convex downward; the gradient of T is discontinuous at the phase transition plane ($T = T_m = 600$ K); as x increases T decreases. In this case the thickness of the amorphous region δ_m is 0.64 mm, which is relatively thin. In the figure, numerical results calculated by the finite difference method (FDM) are also shown. The numerical calculation for each of the amorphous and the crystalline regions was conducted separately by the finite difference method. For the amorphous region, the transient heat transfer problem of Eq. (3.1-1b) with $dS/dt = V$ (a constant value) was solved by numerical calculation with the FDM. The initial condition is $T(x, 0) = T_0$, and the boundary conditions are $T(0, t) = T_s$ and $T(\delta_m, t) = T_m$, in which δ_m is the solution of Eq. (3.4-5) for $T = T_m$. For the crystalline region, the transient heat transfer problem of Eq. (3.1-1b) in which dS/dt and x are replaced by V and z , respectively, was solved by numerical calculation with the FDM. The initial condition is $T(z, 0) = T_0$, and the boundary conditions are $T(0, t) = T_m$, and $T(L_\infty, t) = T_0$, where L_∞ is set to be 0.05 m as a large value. T_s, T_0, T_m, V, k, C_p and ρ are the same as those for the analytical solution. The time step for the calculation is 0.01 s for both regions. The calculated temperature for the time of 10000 s is adopted for the solution by the FDM for each region. The figure shows that the analytical and numerical solutions agree well.

As for the validity of the analytical solution, each analytical solution in Section 3.3.1 to 3.3.8 and Section 3.4.1 to 3.4.6 was calculated and was compared with the solution by FDM. The analytical and numerical solutions agreed well.

Temperature distributions of the Teflon material by the two-phase model in the quasi-steady state condition for $V = 7.5545 \times 10^{-5}$ m/s with various T_s (900, 950, 1000 K) and for $T_s = 950$ K with various V ($0.5 \times 7.5545 \times 10^{-5}$, 7.5545×10^{-5} , $2 \times 7.5545 \times 10^{-5}$ m/s) were calculated by using the analytical solutions of Eqs. (3.4-5) and (3.3-3). Calculated results are shown in Figs. 7 and 8. In this two-phase model, for $T > 600$ K the material is amorphous, and for $T < 600$ K the material is crystalline. In Fig. 7, it is seen that as T_s increases, the temperature increases and the thickness of the amorphous phase also increases. In Fig. 8, it is seen that as V increases, the temperature decreases and the

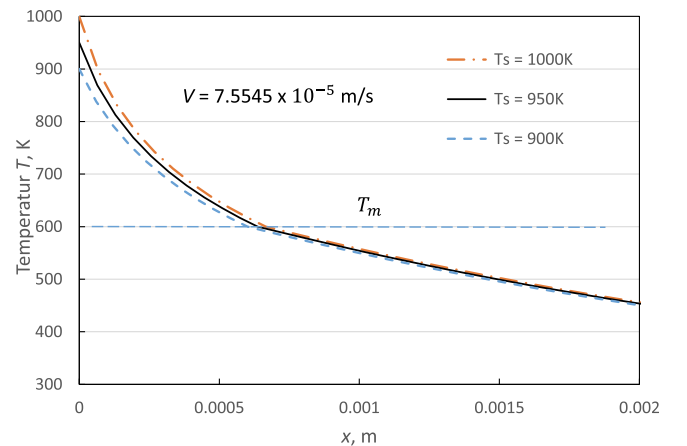


Fig. 7. Effect of T_s upon the temperature distribution (Quasi-steady two-phase model).

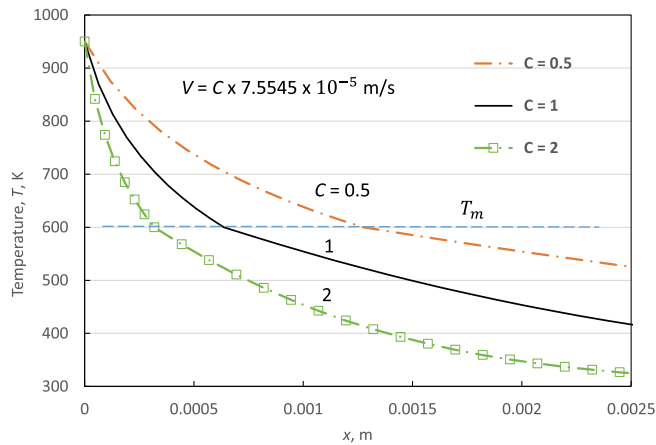


Fig. 8. Effect of V upon the temperature distribution (Quasi-steady two-phase model).

thickness of the amorphous phase also decreases. Within the parameter range of T_s and V in these figures, the thickness of the amorphous phase is small, and the effect of V upon the temperature distribution and the thickness of the amorphous phase is larger than that of T_s .

4. Conclusions

Exact quasi-steady solution of temperature distribution for one-dimensional non-charring ablation in a semi-infinite material with temperature-dependent thermo-physical properties (thermal conductivity, specific heat, and density) is obtained analytically, in which the surface temperature and the surface recession velocity are assumed to be constant. The analytical solutions for the cases in which thermo-physical properties are expressed by low degree polynomial functions of temperature are also obtained. Two kinds of non-charring ablation models are considered. One is a single phase model composed of a single phase (the crystalline (solid) phase), in which the surface recedes due to ablation. The other is a two-phase model composed of two phases (the crystalline (solid) and the amorphous (gel) phases) such as Teflon, in which the surface of the amorphous region recedes. At the boundary between the crystalline and the amorphous regions (phase transition plane), phase transition from the crystalline to the amorphous phase occurs.

The analytical solution in this paper is given in a simple expression. Especially when the thermo-physical properties are expressed by low degree polynomial functions of temperature, the solution is found to be simple and easy to calculate. The calculated results of the analytical solution obtained in this paper agreed well with those of the numerical solution by the finite difference method.

The ablation behavior of Teflon was calculated by using the analytical solutions of single phase and two-phase models with T_s and V varied. It is found that for both single phase and two-phase models, the temperature increases as T_s increases and the temperature decreases as V increases. For the two-phase model, the thickness of the amorphous phase increases as T_s increases and the thickness of the amorphous phase decreases as V increases. Within the parameter range calculated in this paper, the effect of V upon the temperature is larger than that of T_s .

The obtained simple and exact solution is helpful in understanding the ablation behavior. It allows for rapid estimation of the ablation behavior for engineering purposes. The solution also provides a means

to verify computer solutions obtained by numerical methods.

CRediT authorship contribution statement

Sumio Kato: Writing – original draft, Conceptualization. **Shoichi Matsuda:** Writing – review & editing, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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